EX:7 WRITE THE PROGRAM FOR BREADTH-FIRST SEARCH (BFS)

AIM

To develop a Python program that implements the **Breadth-First Search (BFS)** algorithm to traverse a graph level by level from a given starting node.

ALGORITHM

Step 1: Start the program.

Step 2: Create the graph using an adjacency list.

Step 3: Accept the starting node from the user.

Step 4: Mark the starting node as visited and insert it into a queue.

Step 5: Remove a node from the queue, display it, and visit all its unvisited adjacent nodes by adding them to the queue.

Step 6: Repeat the process until the queue becomes empty, then end the program.

PROGRAM CODE:

```
from collections import deque


graph = {}


n = int(input("Enter number of vertices: "))


for i in range(n):
    vertex = input("Enter vertex: ").strip()
    neighbours = input("Enter neighbours separated by space: ").split()
    graph[vertex] = neighbours


start = input("Enter starting node: ").strip()
```

```
visited = set()
queue = deque([start])

print("\nBFS Traversal:")

while queue:
    vertex = queue.popleft()

    if vertex not in visited:
        print(vertex, end=" ")
        visited.add(vertex)

        for neighbour in graph.get(vertex, []):
            if neighbour not in visited:
                queue.append(neighbour)
```

OUTPUT:

```
===== RESTART: C:/Users/Hp/AppData/Local/Prograr
Enter number of vertices: 4
Enter vertex: a
Enter neighbours separated by space: b c
Enter vertex: b
Enter neighbours separated by space: a d
Enter vertex: c
Enter neighbours separated by space: a
Enter vertex: d
Enter neighbours separated by space: b
Enter starting node: a

Graph:
    A
   / \
  B   C
   \ /
    D

BFS Traversal:
a b c d
```

RESULT:

The Python program successfully implements the Breadth-First Search (BFS) algorithm to traverse the graph.The nodes are visited and displayed in **level-by-level (breadth-first)** order starting from the given source node.